# Lecture 28.13: Variance Reduction

## 📌 Topic Overview

In classification, we used Information Gain (reduction in entropy) to choose the best split. In regression trees, the equivalent metric is **Variance Reduction (VR)**. In this lecture, we study the formula and examine the step-by-step blackboard comparison of split candidates.

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## 🎯 What is Variance Reduction?

> **Variance Reduction measures how much the target variance drops after splitting a node into child subsets.**

$$\text{Variance Reduction} = \text{Variance of Parent} - \text{Weighted Variance of Children}$$

### Mathematical Formula:

$$VR = V_S - \left( \frac{|S_L|}{|S|} V_L + \frac{|S_R|}{|S|} V_R \right)$$

Where:

- $V_S$ = Variance of the parent node.
- $V_L, V_R$ = Variance of the left and right child subsets.
- $\frac{|S_L|}{|S|}, \frac{|S_R|}{|S|}$ = Proportion of samples going to the left and right children.

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## 🧮 Blackboard Numerical Example: Comparing Splits

Dataset: $x = [1, 2, 3, 4], y = [2, 4, 6, 10]$. From the previous lecture, parent variance is:

$$V_S = 8.75$$

### Case 1: Split at $x \leq 2$

- **Left Child ($S_L$)**: Contains samples with $x \in \{1, 2\} \overset{\circ}{\rightarrow} y \in \{2, 4\}$.
- Mean: $\bar{y}_L = \frac{2+4}{2} = 3$.
- Variance: $V_L = \frac{(2-3)^2+(4-3)^2}{2} = \frac{1+1}{2} = 1.0$.
- **Right Child ($S_R$)**: Contains samples with $x \in \{3, 4\} \overset{\circ}{\rightarrow} y \in \{6, 10\}$.
- Mean: $\bar{y}_R = \frac{6+10}{2} = 8$.
- Variance: $V_R = \frac{(6-8)^2+(10-8)^2}{2} = \frac{4+4}{2} = 4.0$.

**Weighted Variance of Children:**

$$\text{Weighted Child Variance} = \frac{2}{4}(1.0) + \frac{2}{4}(4.0) = 0.5 + 2.0 = 2.5$$

**Variance Reduction for Case 1:**

$$VR_1 = V_S - 2.5 = 8.75 - 2.5 = 6.25$$

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### Case 2: Split at $x \leq 3$

- **Left Child**: $y \in \{2, 4, 6\} \overset{\circ}{\rightarrow} \bar{y} = 4, V_L = \frac{4+0+4}{3} = \frac{8}{3} \approx 2.67$.
- **Right Child**: $y \in \{10\} \overset{\circ}{\rightarrow} V_R = 0$ (single point $\overset{\circ}{\rightarrow}$ zero variance).
- Weighted Child Variance: $\frac{3}{4}(2.67) + \frac{1}{4}(0) = 2.0$.
- Variance Reduction for Case 2:

$$VR_2 = 8.75 - 2.0 = 6.75$$

### Selection Decision:

Since $VR_2(6.75) > VR_1(6.25)$, the algorithm selects split candidate 2 ($x \leq 3$) because it achieves a greater reduction in variance!

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## 💻 Python Implementation

```
import numpy as np
```

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def calculate_variance_reduction(y_parent, y_left, y_right):
    """
    Calculates Variance Reduction achieved by splitting continuous target y into t
    Formula: VR = Var(Parent) - [ (|Left|/|Parent|)*Var(Left) + (|Right|/|Parent|)
    """
    n_p = len(y_parent)
    n_l = len(y_left)
    n_r = len(y_right)

    v_parent = np.var(y_parent)
    v_left = np.var(y_left)
    v_right = np.var(y_right)

    weighted_child_var = (n_l / n_p) * v_left + (n_r / n_p) * v_right
    reduction = v_parent - weighted_child_var
    return reduction

# --- Compare Candidate Splits on Video Example ---
y = np.array([2, 4, 6, 10])

# Split 1: x <= 2 -> Left [2, 4], Right [6, 10]
vr_split1 = calculate_variance_reduction(y, np.array([2, 4]), np.array([6, 10]))

# Split 2: x <= 3 -> Left [2, 4, 6], Right [10]
vr_split2 = calculate_variance_reduction(y, np.array([2, 4, 6]), np.array([10]))

print(f"Variance Reduction (Split 1: x <= 2): {vr_split1:.3f}")
print(f"Variance Reduction (Split 2: x <= 3): {vr_split2:.3f}")
print("Best Split:", "Split 2 (x <= 3)" if vr_split2 > vr_split1 else "Split 1 (x
```

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## 🔑 Key Takeaways

1. Variance Reduction is the regression equivalent of Information Gain.
2. The tree evaluates all potential split thresholds and picks the one that maximizes Variance Reduction.
3. A higher Variance Reduction means the child subsets are more tightly clustered around their respective means.